

Pedro Beber de Queiroz Vidal

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LLM agents and computer vision researcher, designing multi-agent architectures at Unicamp's HIAAC with a PROEX/Unicamp graduate scholarship. Eleven publications in four years, three as first author, and podium finishes in four international challenges, including **1st place** in a sub-task of the FRCSyn Challenge at WACV 2024.

EDUCATION

University of Campinas (Unicamp)

MSc in Computer Science — PROEX/Unicamp scholarship holder

Thesis: architectures for LLM multi-agent systems, integrating dynamic memory retrieval and inter-agent communication.

Campinas, SP, Brazil

2026 — Mar 2028 (expected)

Federal University of Paraná (UFPR)

BSc in Computer Science

Thesis on synthetic facial data, extended into a survey submitted to IEEE Access.

Curitiba, PR, Brazil

2020 — Jul 2025

Federal Institute of Paraná (IFPR)

Technical Diploma in Informatics

Irati, PR, Brazil

2016 — 2019

RESEARCH EXPERIENCE

HIAAC, University of Campinas (Unicamp)

Graduate Researcher

Campinas, SP, Brazil

Aug 2026 — Present

- Design a multi-agent architecture for LLM agents that couples dynamic memory retrieval with an optimized inter-agent communication protocol, evaluated against state-of-the-art agentic benchmarks.
- Benchmark reasoning and orchestration strategies — ReAct, chain-of-thought, in-context learning, task decomposition — to isolate which choices actually move task success under controlled conditions.
- Authored a standalone literature review of LLM multi-agent systems, mapping communication topologies, memory mechanisms, and evaluation practice across the field.

BOVIFOCR, UFPR-UNICO Partnership

Undergraduate Researcher, Computer Vision

Curitiba, PR, Brazil

Oct 2021 — Mar 2025

- Built the submission that took **1st place** in Sub-Task 2.1 of the FRCSyn Challenge at WACV 2024, and the SDFR entries that placed 3rd in both tasks at FG 2024.
- Trained and evaluated face recognition models on synthetic identity datasets of millions of images, adapting InsightFace and running the pipeline end to end from data processing to benchmark reporting.
- Quantified the impact of synthetic training data across eight standard benchmarks, identifying which dataset characteristics predict recognition performance; published at SIBGRAPI 2024 and extended into a survey submitted to IEEE Access.
- Authored the group's masked face recognition benchmark (SIBGRAPI 2022) and represented it at the Occluded Face Recognition competition at IJCB 2022.
- Built reproducible benchmark protocols over standardized datasets to separate genuine model gains from training-data effects, tracking every run in Weights & Biases.
- Collected liveness samples in the field, assisting elderly participants unfamiliar with the technology, and drove adoption of the collection app on social networks.

- Improved occluded face verification by extending Earth Mover's Distance as the matching metric, validating the gain through False Non-Match Rate at fixed False Match Rate; published at SIBGRAPI 2023.
- Selected for a summer school on innovation and startups with international students, followed by an internship at the university's computer science research institute.
- Proposed a financial planning application for university students covering tuition, living costs, and scholarship tracking.

HONORS AND COMPETITIONS

FRCSyn Challenge, 1st Edition, WACV 2024 — **1st place**, Sub-Task 2.1; 3rd, 4th and 4th in Sub-Tasks 1.1, 1.2 and 2.2. 2023

SDFR Competition, FG 2024 — 3rd place in both tasks of Synthetic Data for Face Recognition. 2024 — 2025

FRCSyn Challenge, 2nd Edition, CVPR 2024 — 6th place, Sub-Task 2.1. 2024

PROEX/Unicamp graduate scholarship — graduate research funding. 2026

OBMEP, Brazilian Mathematics Olympiad — Honorable Mention. 2019

PROJECTS

Federated Learning for Privacy-Preserving Facial Recognition — [code](#) 2026 · Unicamp
2D face recognition trained without centralizing biometric data, combining federated clients with synthetic facial data to hold accuracy under non-IID splits, limited client compute, and scarce local data.

Comparative Evaluation of Synthetic Facial Datasets — [thesis](#) 2025 · UFPR
Trained face recognition models on multiple synthetic datasets and scored them across eight benchmarks, isolating the dataset characteristics and quality trade-offs behind stronger and weaker performance.

Training Pipeline for Synthetic-Data Competitions — [code](#) 2024 · UFPR
The pipeline behind the SDFR and FRCSyn submissions: InsightFace adapted for training and evaluation on million-image synthetic identity datasets, with results organized for comparison across datasets, model configurations, and benchmark protocols.

Robotics Team, Irathunder Group 2017 — 2019 · IFPR
Programmed and built LEGO robots for rescue, sumo, and line-following events; competed in two editions of the Brazilian Robotics Olympiad (OBR).

TECHNICAL SKILLS

LLMs, agents	Multi-agent systems, inter-agent communication, dynamic memory retrieval, ReAct, chain-of-thought, in-context learning, task decomposition, agentic benchmarking.
Machine learning	PyTorch, TensorFlow, MXNet, Keras, OpenCV, InsightFace, Flower; generative models, face recognition, liveness detection, knowledge distillation, LoRA and parameter-efficient fine-tuning, Federated Learning.
Data	NumPy, Pandas, Scikit-learn, Matplotlib; dataset validation, data cleaning, exploratory analysis, metric computation, statistical comparison of results.
Engineering	Python (advanced), Git, Linux, shell scripting, Conda, Jupyter, Weights & Biases; reproducible experiment design, distributed training, large-scale dataset processing.
Languages	Portuguese (native), English (fluent).

PUBLICATIONS

FIRST AUTHOR

1. **Vidal, Pedro**, Bernardo Biesseck, et al. (2024). "A Comparative Study on Synthetic Facial Data Generation Techniques for Face Recognition". In: *2024 37th SIBGRAPI Conference on Graphics, Patterns and Images*. sol.sbc.org.br
2. **Vidal, Pedro**, Henry Chu, et al. (2023). "Extending Earth Mover's Distance to Occluded Face Verification". In: *2023 36th SIBGRAPI Conference on Graphics, Patterns and Images*. IEEE, pp. 49–54. sibgrapi.sid.inpe.br
3. **Vidal, Pedro**, Roger Leitzke Granada, et al. (2022). "A Benchmark on Masked Face Recognition". In: *Proceedings... Conference on Graphics, Patterns and Images, 35 (SIBGRAPI)*. urlib.net
4. **Vidal, Pedro Beber** (2024). *A Comparative Study on Synthetic Facial Data Generation Techniques for Face Recognition*. Undergraduate thesis, UFPR, Curitiba. bit.ly/3V03Efp

CO-AUTHORED

5. Melzi, Pietro et al. (2024). "FRCSyn-onGoing: Benchmarking and comprehensive evaluation of real and synthetic data to improve face recognition systems". In: *Information Fusion* 107, 102322. doi.org
6. DeAndres-Tame, Ivan et al. (2025). "Second FRCSyn-onGoing: Winning solutions and post-challenge analysis to improve face recognition with synthetic data". In: *Information Fusion* 120, 103099. doi.org
7. Biesseck, Bernardo, **Pedro Vidal**, et al. (2024). "TCDiff: Triple Condition Diffusion Model with 3D Constraints for Stylizing Synthetic Faces". In: *2024 37th SIBGRAPI Conference on Graphics, Patterns and Images*, pp. 1–6. scholar.google.com
8. Biesseck, Bernardo, **Pedro Vidal**, et al. (2023). "3D Face Reconstruction as Complementary Data to Enhance Face Recognition". In: *2023 36th SIBGRAPI Conference on Graphics, Patterns and Images*, pp. 61–66. scholar.google.com
9. Shahreza, Hatef Otroshi et al. (2024). "SDFR: Synthetic Data for Face Recognition Competition". In: *2024 IEEE 18th International Conference on Automatic Face and Gesture Recognition (FG)*. IEEE, pp. 1–9. ieeexplore.ieee.org
10. DeAndres-Tame, Ivan et al. (2024). "FRCSyn Challenge at CVPR 2024: Face Recognition Challenge in the Era of Synthetic Data". In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 3173–3183. openaccess.thecvf.com
11. Melzi, Pietro et al. (2024). "FRCSyn Challenge at WACV 2024: Face Recognition Challenge in the Era of Synthetic Data". In: *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pp. 892–901. openaccess.thecvf.com
12. Neto, Pedro C et al. (2022). "OCFR 2022: Competition on Occluded Face Recognition from Synthetically Generated Structure-Aware Occlusions". In: *International Joint Conference on Biometrics 2022*. ieeexplore.ieee.org